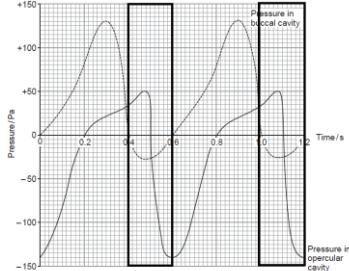
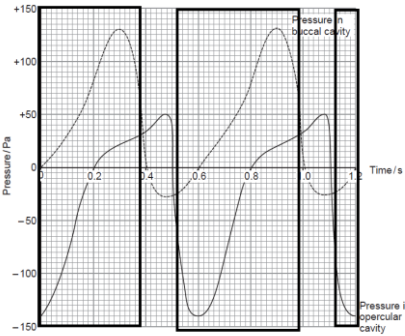


Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			Any four (x1) from: A. Mouth opens and <u>floor</u> of buccal cavity lowered (1) B. Increase volume and decreases pressure (in buccal cavity) (1) C. Water enters (the buccal cavity) (1) D. Mouth closes and <u>floor</u> of buccal cavity raised / opercular cavity expands (1) E. Decreases volume and increases pressure (in buccal cavity) / increases volume and decreases pressure in the opercular cavity (1) F. Water is {forced / pushed / moves down a pressure gradient} over the gills (1) MP C can only be awarded in context of MP A or B MP F can only be awarded in context of MP D or E	4			4		
	(b)	(i)		190 Pa ignore – sign		1		1	1	
		(ii)	I	B in any region of the graph when the buccal pressure is below 0pa 			1	1		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
			II	G in any region of the graph when the buccal pressure is greater than the opercular pressure. NOT where lines cross 			1	1		
	(c)	(i)		(secondary) lamellae / gill plates (accept lamella) NOT filaments		1		1		
		(ii)		The gills have {many / flattened / thin} {lamellae / gill plates / filaments} (1) Which provide a large surface area (1) ignore SA to volume ratio OR {lamellae / gill plates / filaments} are {thin / flattened} (1) Which provide a short diffusion pathway (1)	2			2		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		- The concentration gradient is maintained / equilibrium is never reached (1) - Oxygen can <u>diffuse</u> into the blood across the <u>whole of the {gill plate / (secondary) lamellae}</u> (1) ignore reference to time <u>More</u> oxygen is absorbed into the blood / blood achieves a <u>higher</u> oxygen saturation / it reaches 80% saturation as compared to 50% (with parallel flow) (1)	3			3		
	(d)			Any two (x1) from {Lungs / Gulping air} enables <u>oxygen</u> to be obtained (when there is insufficient dissolved oxygen in the water) (1) Many capillaries maintain a steep concentration gradient (1) Many (small) air sacs provide a large surface area (1)			2	2		
	(e)			Lungfish and mammals share a common ancestor / shows divergent evolution (1) Reject mammals evolved from lungfish			1	1		
				Question 5 total	9	2	5	16	1	0